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Machine Learning

The Art and Science of Algorithms that Make Sense of Data

Dr. Mitra Baratchi

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Leiden Institute of Advanced Computer Science - Leiden University

Recap of binary classification

- Confusion matrices and ROC curves
- Scoring and ranking
- Class probability estimation

Session 3: Beyond binary classification ¹

Reading material

→ Chapter 3, Machine learning: the art and science of algorithms that make sense of data

¹The slides partially include material based on the slides accompanying the book: ***Machine learning: the art and science of algorithms that make sense of data*** Cambridge University Press, 2012.

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1. Multi-class classification
 - Creating multi-class classifiers
 - Evaluating multi-class classifier
2. Regression
3. Unsupervised and descriptive learning
 - Predictive and descriptive clustering
 - Sub-group discovery

Multi-class classification

Beyond binary classification

Many concepts are not binary:

Beyond binary classification

Many concepts are not binary:

- Healthy versus ill (binary), or disease type diagnosis (multi-class)

Beyond binary classification

Many concepts are not binary:

- Healthy versus ill (binary), or disease type diagnosis (multi-class)
- Segmentation of text based on the same versus a different topic (binary), or topic classification (multi-class)

Multi-class classification

Multi-class classification

Multi-class classification based on the label space $\mathcal{L} = \mathcal{C}^2$ and output space $\mathcal{Y} = \mathcal{C}$ considers learning an approximation $\hat{c} : \mathcal{X} \rightarrow \mathcal{C}^3$ to the true labelling function c .

² $\mathcal{C} = \{C_1, C_2, \dots, C_k\}$

³ $\mathcal{X}$ denotes the instance space

Beyond binary classification

- How can we build multi-class models?
- How can we evaluate multi-class performance?

Example



Emergency room example

In an emergency room for each patient, a report with 15 variables is made (e.g., blood pressure, age, gender, ...). We want to know where the patient should be transferred to next (ICU, CCU, regular ward):

Example



Emergency room example

In an emergency room for each patient, a report with 15 variables is made (e.g., blood pressure, age, gender, ...). We want to know where the patient should be transferred to next (ICU, CCU, regular ward):

- Features: the 15 variables in the report
- Class labels: ICU, CCU, Regular Ward (RW)

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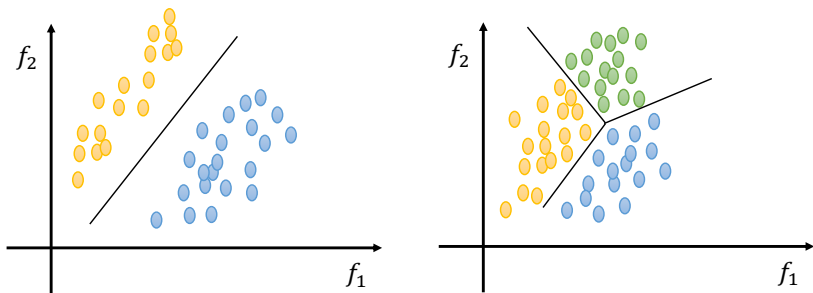
Multi-class classification

Types of classification algorithms:

- Inherently non-binary (e.g., decision trees)
- Inherently binary (e.g., Support Vector Machines (SVM))

→ Binary classifiers can be turned into multi-class classifiers by a variety of strategies

Binary versus multi-class classification



Binary versus multi-class classifier

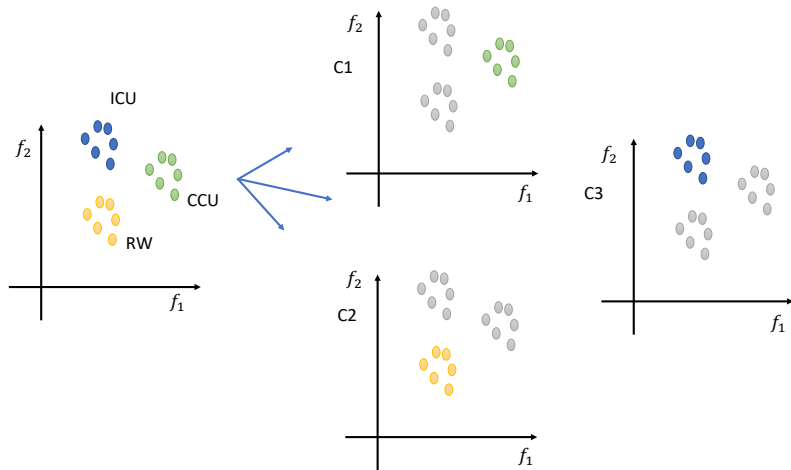
How can we turn the binary classifier on the left to the multi-class classifier on the right?

Turning a binary classifier into a multi-class classifier

Three main approaches for turning binary classifiers into multi-class classifiers:

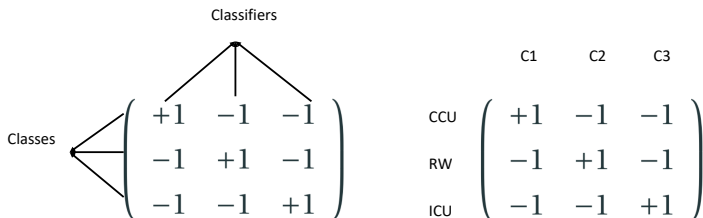
- *One versus rest*
- *One versus one*

One versus rest



One versus rest classifiers

Code matrix



(left) A code matrix (right) code matrix of the emergency room example

Output code matrix

Output code matrix is a k (number of classes) by l (classifiers) matrix whose entries are +1, -1, or 0. Each entry $c_{i,j}$ represents how class i has been encoded to a binary class for the j classification task. +1 is the positive class, -1 is the negative class, and 0 is no prediction.

One versus rest

$$\begin{pmatrix} +1 & -1 & -1 \\ -1 & +1 & -1 \\ -1 & -1 & +1 \end{pmatrix}$$

$$\begin{pmatrix} +1 & 0 \\ -1 & +1 \\ -1 & -1 \end{pmatrix}$$

The code matrices for the unordered and ordered version of the one-versus-rest schemes

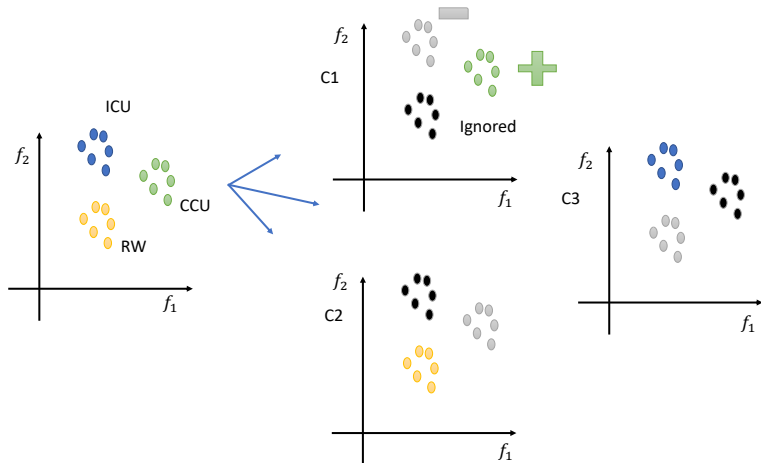
One versus rest

The process of one versus rest classification:

- **Learning:** train k or $k - 1$ separate classifiers (k is the number classes)
- **Inference:** use all of the classifiers and form a code word based on the output of the classifier. Next, compare the code word against all the rows and find the nearest row in the code matrix.

→ You will get an incorrect classification if any of these classifiers makes a misclassification.

One versus one or all pairs



One versus one classifiers

Output code matrix

$$\begin{pmatrix} +1 & +1 & 0 \\ -1 & 0 & +1 \\ 0 & -1 & -1 \end{pmatrix}$$

$$\begin{pmatrix} +1 & -1 & +1 & -1 & 0 & 0 \\ -1 & +1 & 0 & 0 & +1 & -1 \\ 0 & 0 & -1 & +1 & -1 & +1 \end{pmatrix}$$

Code matrix C describing the two ways to transform a three-class task by means of one-versus-one.

→ If a binary classifier treats the classes asymmetrically, it makes more sense to train two classifiers for each pair.

One versus one

Question

How many symmetric and asymmetric binary classifiers are needed to create a one versus one classifier of k classes?

- 3 classes ->

One versus one

Question

How many symmetric and asymmetric binary classifiers are needed to create a one versus one classifier of k classes?

- 3 classes -> 3 and 6
- 4 classes ->

One versus one

Question

How many symmetric and asymmetric binary classifiers are needed to create a one versus one classifier of k classes?

- 3 classes -> 3 and 6
- 4 classes -> 6 and 12
- ...

Number of classifiers

$k(k-1)/2$ binary classifiers are needed to train a symmetric classifier with k number of classes and $k(k-1)$ classifiers for the asymmetric case.

One versus one

The process of one versus one classification:

- **Training:** train separate classifiers for each pair of classes
- **Inference:** use all classifications for a code word based on the output of the classifier. Next, compare the code word against all rows and find the nearest row in the code matrix. Take a voting scheme when distances are not unique.

→ More accurate but slower inference

Comparing the complexities

	Training	Inference
One versus all	$\mathcal{O}(km^\alpha)$	$\mathcal{O}(k\beta)$
One versus one	$\mathcal{O}(k^2(\frac{m}{k})^\alpha)$	$\mathcal{O}(k^2\beta)$

- k is the number of classes
- $\mathcal{O}(m^\alpha)$ is the complexity of training on m instances
- $\mathcal{O}(\beta)$ is the complexity of inference on one instance

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Performance of multi-class classifier

Consider the following three-class confusion matrix (plus marginals):

		<i>Predicted</i>			
<i>Actual</i>	ICU	15	2	3	20
	CCU	7	15	8	30
	RW	2	3	45	50
		24	20	56	100

How do we extend the previous metrics to it?

- The accuracy
- Precision and recall

Accuracy of a multi-class classifier

		<i>Predicted</i>			
<i>Actual</i>	15	2	3	20	
	7	15	8	30	
	2	3	45	50	
	24	20	56	100	

- The accuracy of the classifier: $(15 + 15 + 45)/100 = 0.75$.

Precision and recall of a multi-class classifier

		<i>Predicted</i>			
<i>Actual</i>	15	2	3	20	
	7	15	8	30	
	2	3	45	50	
	24	20	56	100	

Precision and recall of a multi-class classifier

		<i>Predicted</i>			
<i>Actual</i>	15	2	3	20	
	7	15	8	30	
	2	3	45	50	
	24	20	56	100	

Per-class precision and recall:

- First class $\rightarrow prec = 15/24 = 0.63, rec = 15/20 = 0.75$
- Second class $\rightarrow prec = 15/20 = 0.75, rec = 15/30 = 0.50$
- Third class $\rightarrow prec = 45/56 = 0.80, rec = 45/50 = 0.90$

Precision and recall for the whole classifier (weighted average based on the proportions) $\rightarrow Prec = 0.20 \cdot 0.63 + 0.30 \cdot 0.75 + 0.50 \cdot 0.80 = 0.75$.

Precision and recall of a multi-class classifier

		<i>Predicted</i>			
<i>Actual</i>	15	2	3	20	
	7	15	8	30	
	2	3	45	50	
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Precision and recall of a multi-class classifier

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	7	15	8		30
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	24	20	56		100

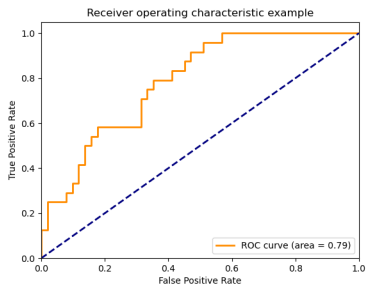
Precision and recall numbers for each pair of classes →

- First and third class → $prec = 15/17 = 0.88$, $rec = 15/18 = 0.83$
- Third and first class → $prec = 45/48 = 0.94$, $rec = 45/47 = 0.96$

ROC and AUC

How can we know how good a classifier can be when setting different thresholds based on these scores/probabilities? → with binary classifiers we had the ROC curve and AUC indicator that considered the proportion of correctly ranked positive–negative pairs.

ROC curve of a binary classifier



ROC curve for a binary classifier

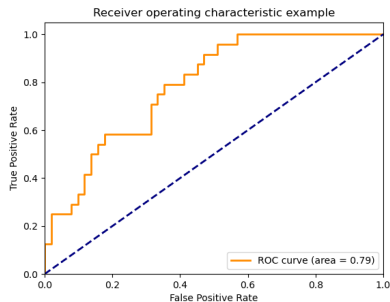
Image source: https://scikit-learn.org/stable/auto_examples/model_selection/plot_roc.html

ROC for multi-class classification

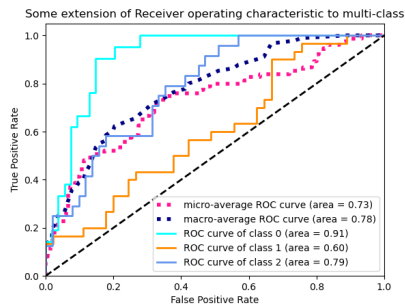
By restricting the attention to each individual class, we can calculate an average ROC curve for more classes (micro-average, macro-average)

- *Macro-average* will compute the metric independently for each class and then take the average (hence treating all classes equally). (e.g., macro sensitivity $\rightarrow \frac{TPR_1 + TPR_2 + TPR_3}{3}$)
- *Micro-average* will aggregate the contributions of all classes to compute the average metric treating each data instance equally (e.g., micro true positive rate (TPR) $\rightarrow \frac{TP_1 + TP_2 + TP_3}{TP_1 + TP_2 + TP_3 + FN_1 + FN_2 + FN_3}$)

ROC curve of a multi-class classifier



ROC curve of a binary classifier



ROC curve of a multi-class classifier

Image source: https://scikit-learn.org/stable/auto_examples/model_selection/plot_roc.html

Regression

Regression

Regression

A *regressor* is a function estimator $\hat{f} : \mathcal{X} \rightarrow \mathbb{R}$ based on the output space $\mathcal{Y} = \mathbb{R}$ that approximates to the true labeling function f using labeled examples $(x_i, f(x_i))$ from the label space $\mathcal{L} = \mathbb{R}$.

- Supervised learning
- A much higher resolution target compared to classification

Line fitting example

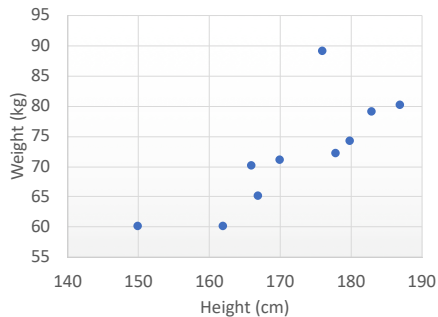
Consider the following set of points:

x (Height (cm))	y (weight (kg))
150	60
161	60
167	65
170	71
178	72

→ We want to estimate y by means of a polynomial in x .

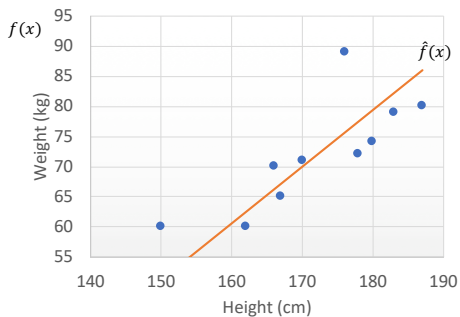
→ It is reasonable to assume that the examples are noisy and that the estimator is only intended to capture the general trend or shape of the function.

Regression



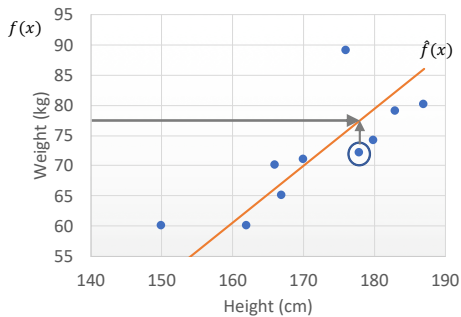
Scatter plot of height versus weight

Regression



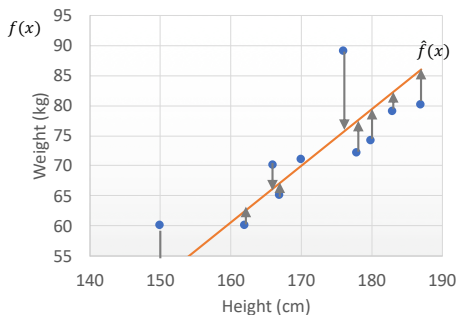
Demonstrating linear regression

Regression



Scatter plot of height versus weight

Residuals



Scatter plot of height versus weight

Regression loss

Regression models are evaluated by applying a loss function to the *residuals* $= f(x) - \hat{f}(x)$.

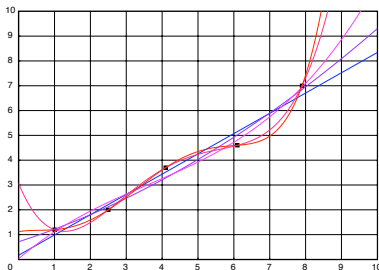
Polynomials

An n -degree polynomial has $n + 1$ parameters:

- $y = a \cdot x + b \rightarrow$ two parameters (in linear regression a is called the slope and b is called the intercept)
- $y = a \cdot x^2 + b \cdot x + c \rightarrow$ three parameters

What happens if we use a polynomial with a higher degree?

Fitting polynomials to data



Polynomials of different degrees fitted to a set of five points. From bottom to top in the top right-hand corner: degree 1 (straight line), degree 2 (parabola), degree 3, degree 4 (which is the lowest degree able to fit the points exactly), degree 5.

Overfitting

- An n -degree polynomial has $n + 1$ parameters:
 - $y = a \cdot x + b \rightarrow$ degree 1, two parameters
- The models that can fit all the training points are the models with more parameters. $\rightarrow$ trying to memorise the training set will lead to *overfitting*.

Overfitting in regression

To avoid overfitting, the number of parameters estimated from the data must be *considerably less* than the number of data points.

Overfitting

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Overfitting in regression

To avoid overfitting, the number of parameters estimated from the data must be *considerably less* than the number of data points.

$\rightarrow$ Does this mean that the linear model is the best?

Bias and variance

Two things we aim for:

- Lower error (i.e., smaller residuals)
- Generalisability

Bias–variance dilemma

Bias–variance dilemma: a low-complexity model suffers less from variability due to random variations in the training data, but may introduce a systematic *bias* that even large amounts of training data can't resolve; on the other hand, a high-complexity model eliminates such bias but can suffer non-systematic errors due to *variance*.

Bias and variance

Bias-variance dilemma

We can make this a bit more precise by noting that the expected squared loss on a training example x can be decomposed as follows:

$$\mathbb{E} \left[(f(x) - \hat{f}(x))^2 \right] = (f(x) - \mathbb{E} [\hat{f}(x)])^2 + \mathbb{E} \left[(\hat{f}(x) - \mathbb{E} [\hat{f}(x)])^2 \right]$$

The first term on the right is zero if these function estimators get it right on average; otherwise, the learning algorithm exhibits a systematic *bias* of some kind. The second term quantifies the *variance* in the function estimates $\hat{f}(x)$ as a result of variations in the training set.

Fitting polynomials to data

See the demo:

https://colab.research.google.com/drive/11E192pQFxxwh_IXQ-epRtyFZOWKmeuO1n?usp=sharing

Unsupervised and descriptive learning

Overall categorisation of ML tasks

Reminder from Session 1:

	<i>Predictive model</i>	<i>Descriptive model</i>
<i>Supervised learning</i>	Classification, Regression	<i>Subgroup discovery</i>
<i>Unsupervised learning</i>	<i>Predictive clustering</i>	<i>Descriptive clustering, Association rule discovery</i>

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Predictive and descriptive clustering

Predictive view of clustering

So we could define a ‘clusterer’ in the same way as a classifier, namely as a mapping $\hat{q} : \mathcal{X} \rightarrow \mathcal{C}$, where $\mathcal{C} = \{C_1, C_2, \dots, C_k\}$ is a set of new labels.

Descriptive view of clustering

A descriptive clustering model learned from given data $D \subseteq \mathcal{X}$ would be a mapping $\hat{q} : D \rightarrow \mathcal{C}$ (D rather than $\mathcal{X}$).

Labels only show the difference between clusters...

Descriptive learning

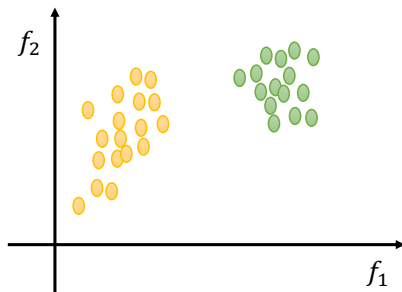
- In descriptive learning the task is to come up with a description of the data
- In descriptive learning it makes no sense to employ a separate training set to produce the descriptive model, as we want the model to describe our actual data rather than some hold-out set.
- It is more likely to have ground truth to evaluate predictive clustering while this is unlikely for descriptive clustering tasks.

Distance-based clustering

Distance-based clustering

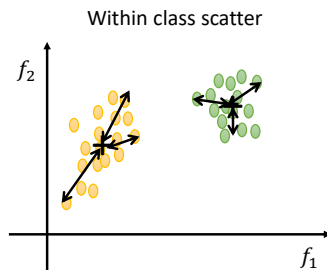
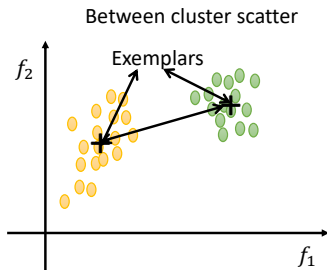
Most distance-based clustering methods depend on the possibility of defining a 'centre of mass' or *exemplar* for an arbitrary set of instances, such that the exemplar minimises some distance-related quantity over all instances in the set, called its *scatter*. A good clustering is then one whereby the scatter is summed over each cluster – the *within-cluster scatter* – is much smaller than the scatter of the entire data set.

Scatter



Instances within two different clusters

Predictive clustering



(left) Between cluster scatter (right) Within class scatter

Evaluating clustering performance

How can we assess clustering performance?

- With ground truth
- Without ground truth

Evaluating clustering (with ground truth)

Suppose we have five test instances that we think should be clustered as $\{e1, e2\}, \{e3, e4, e5\}$

Evaluating clustering (with ground truth)

Suppose we have five test instances that we think should be clustered as $\{e1, e2\}, \{e3, e4, e5\}$

We want to evaluate the clustering that clusters these examples as $\{e1, e2, e3\}, \{e4, e5\}$

Evaluating clustering (with ground truth)

Suppose we have five test instances that we think should be clustered as $\{e1, e2\}, \{e3, e4, e5\}$

We want to evaluate the clustering that clusters these examples as $\{e1, e2, e3\}, \{e4, e5\}$

	<i>Are together</i>	<i>Are not together</i>	
<i>Should be together</i>	2	2	4
<i>Should not be together</i>	2	4	6
	4	6	10

Evaluating clustering performance (with ground truth)

Suppose we have five test instances that we think should be clustered as $\{e1, e2\}, \{e3, e4, e5\}$

We want to evaluate the clustering that clusters these examples as $\{e1, e2, e3\}, \{e4, e5\}$

	<i>Are together</i>	<i>Are not together</i>	
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<i>Should not be together</i>	2	4	6
	4	6	10

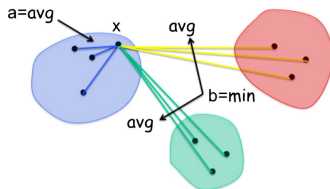
- **Rand-index** (equivalent of accuracy): $6/10$
- **Precision**: $TP/(TP + FP) = 2/4$
- **Recall**: $TP/(TP + FN) = 2/4$
- **F-1** = $\frac{2 \times \text{prec} \times \text{recall}}{\text{prec} + \text{recall}} = 1/2$

Evaluating clustering performance (without ground truth): Silhouette coefficient

Silhouette coefficient

$$s = \frac{b-a}{\max(a,b)}$$

- a: the mean distance between an instance and all other points in the same cluster.
- b: the mean distance between an instance and all other points in the next nearest cluster.



Silhouette coefficient

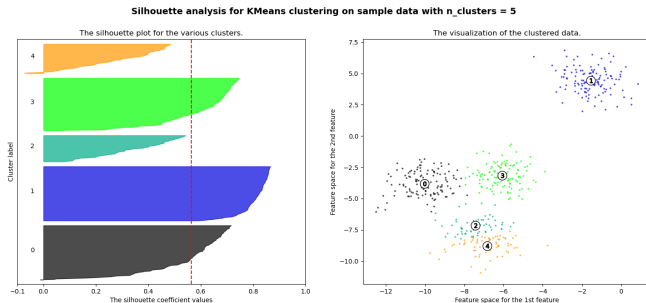
Silhouette coefficient

$$s = \frac{b-a}{\max(a,b)}$$

- a: the mean distance between an instance and all other points in the same cluster.
- b: the mean distance between an instance and all other points in the next nearest cluster.

The silhouette ranges from -1 to +1 → Silhouette coefficients near +1 indicate that the instance is far away from the neighboring clusters. A value of 0 indicates that the instance is on or very close to the decision boundary between two neighboring clusters and negative values indicate that those instances might have been assigned to the wrong cluster.

Silhouette coefficient



Bad performance due to the presence of clusters with below average silhouette coefficients

Image source: https://scikit-learn.org/stable/auto_examples/cluster/plot_kmeans_silhouette_analysis.html

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Subgroup discovery

Subgroup discovery is a supervised learning task but it is different from classification, as it addresses different goals → discovery of interesting population subgroups instead of maximizing classification accuracy of the induced rule set.

- Examples:
 - Detection of risk groups with coronary heart disease, or cancer
 - Finding patterns in traffic accidents

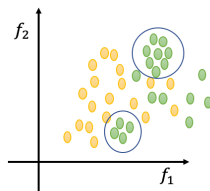
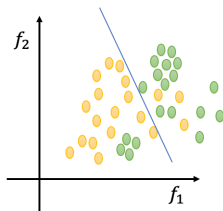
Subgroup discovery

Subgroup discovery

A *subgroup* is a mapping $\hat{g}: D \rightarrow \{true, false\}$ ($D \subseteq \mathcal{X}$) learned from a set of labelled instances $(x_i, l(x_i))$, where $l: \mathcal{X} \rightarrow \mathcal{C}$ is the true labelling function. Note that $\hat{g}$ is the characteristic function of the set $G = \{x \in D | \hat{g}(x) = true\}$ and is called the extension of a subgroup.

- Subgroup models don't try to approximate the labelling function.
- They aim at identifying the population of data that are sufficiently large and statistically unusual.

Subgroup discovery



(left) Classifier (right) Sub-group discovery

Evaluating the performance of subgroup discovery

How can we assess the performance of subgroup discovery?

Evaluating subgroups

	<i>In subgroup</i>	<i>Not in subgroup</i>	
<i>Labelled</i> C_1	g_1	$C_1 - g_1$	C_1
<i>Labelled</i> C_2	g_2	$C_2 - g_2$	C_2
<i>Labelled</i> C_3	g_3	$C_3 - g_3$	C_3
	$ G $	$ D - G $	$ D $

- Measuring the extent to which the class distribution in the left column is different from the class distribution in the row marginals (the right-most column).
- Chi-squared test** $\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$ (O_i is the observed and E_i is the expected value) can evaluate the extent to which each g_i differs from what would be expected on the basis of the marginals C_i and $|G|$.

Lessons learned

- Multi-class classification
 - How to make them? One versus one, one versus rest
 - How to assess their performance? Extending binary evaluation metric and ROC curves
- Regression
 - How to estimate regressor functions using polynomials?
 - Bias-variance dilemma
- Descriptive and predictive clustering
- Subgroup discovery